



x

1

7

x

10

0

1

0

1



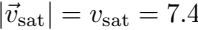




PARADOXES

ESPRESSO

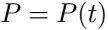








BEAR





THE FIRST TWO



100%







2023-10-21

$$= \underbrace{\frac{p_t^2}{2 \log(2)} \pi \lambda \left(\frac{h \varphi_{\text{RX}}}{\sin^2(\epsilon)} \right)^2}_{k \triangleq} \exp \left\{ - \tau^2 \underbrace{v_{\text{sat}}^2 \sin^4(\epsilon) / (h^2 \varphi_{\text{RX}}^2)}_{D \triangleq} \log(2) / 2 \right\},$$



























Fig 1(a): True $P(t)$, GPR estimate & forecast, and 100-sample MA & forecast

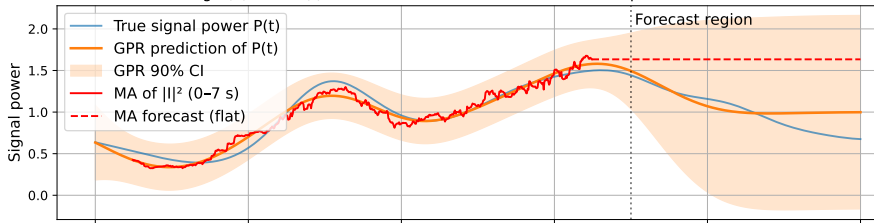
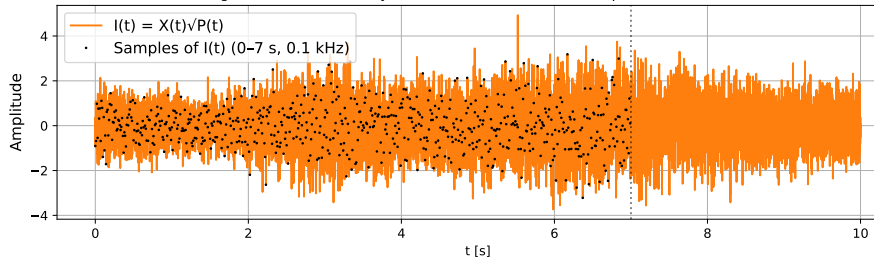


Fig 1(b): Non-stationary Gaussian noise $I(t)$ with sampled observations







1995



expanding world + new world







Qatar Airways









$$S_{RX}(t) = \frac{A}{l(d_0)} \sqrt{\frac{K}{K+1}} e^{-2i\pi\tau_0(t)f_c} e^{-2i\pi\theta} + \sum_{j=1}^{100} \frac{A}{l(d_j) \sqrt{100} \sqrt{K+1}} e^{-2i\pi\tau_j(t)f_c} e^{-2i\pi\theta},$$

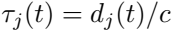


















QWERTYUIOP

$$v = \sqrt{\frac{GM}{h + R_{\oplus}}},$$

WELCOME TO THE









20

—

12

.

100



$$T_c = \frac{1}{8D_g} \approx 10^{-3}$$









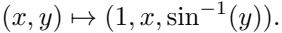


















$$1 = \frac{1}{e^{\frac{\pi^2}{2\sqrt{P}}}}$$





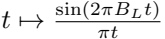
$$] = 1 - \left(1 - e^{-\frac{\pi^2}{2\sqrt{P/N}}} \right)^N,$$





192







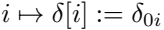
$$t \rightarrow \text{order} \rightarrow \frac{\sin(2\pi B t)}{\pi t}$$





$$\mathbb{Z} \ni i \mapsto \frac{\sin\left(2\pi \frac{B_L}{f_c} i\right)}{\pi i},$$

$$i \mapsto \delta[i] - \frac{\sin(2\pi \frac{B_H}{f_c} i)}{\pi i},$$



গণপ্রজাতন্ত্রী বাংলাদেশ

WAVELENGTH



WINNERS

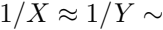
$$\mu = \exp\left(\mu_{\text{LN}} + \frac{\sigma_{\text{LN}}^2}{2}\right)$$

$\sin^2 \theta = 1 - \sin^2 \theta$



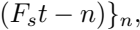






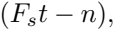
FORGIVENESS





1991-10-1

$$S(t) = \sum_{n=0}^{N-1} x[n]$$













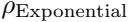


$$H = \begin{pmatrix} -a & -0 & -1 & 0 & 0 & 0 & ar & 0 & r \\ 0 & 0 & 0 & -a & -0 & -1 & 0 & 0 & 0 \\ -c & -b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & -b^2/a & -1 & cr & rb^2/a & r \\ a & 0 & -1 & 0 & 0 & 0 & ar & 0 & -r \\ 0 & 0 & 0 & a & 0 & -1 & 0 & 0 & 0 \\ -c & b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & b^2/a & -1 & -rc & rb^2/ar & -r \end{pmatrix}$$

$$\lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X \geq x)}{x},$$







$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x}$$

$$= \lim_{x \rightarrow \infty} \frac{-x/\theta}{x} = 1/\theta.$$





$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(1 - \gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$\begin{aligned}
 &\stackrel{(a)}{=} \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta))}{x} = \lim_{x \rightarrow \infty} - \frac{\log(x^{k-1} e^{-x/\theta})}{x}
 \end{aligned}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x} = 1/\theta.$$



$$\lim_{x \rightarrow \infty} \frac{\Gamma(s, x)}{x^{s-1} e^{-x}} = 1.$$







$$\begin{aligned}
&= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = - \lim_{x \rightarrow \infty} \frac{\log \operatorname{erfc} \left(\frac{x - \mu}{\sigma \sqrt{2}} \right)}{x} \stackrel{(b)}{\geq} - \lim_{x \rightarrow \infty} \frac{\log \left(\sqrt{\frac{e}{2\pi}} e^{-2 \left(\frac{x - \mu}{\sigma \sqrt{2}} \right)^2} \right)}{x}
\end{aligned}$$

$$\geq -\lim_{x \rightarrow \infty} \frac{\log \left(e^{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2} \right)}{x} = -\lim_{x \rightarrow \infty} \frac{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2}{x} = \frac{1}{\sigma^2} \lim_{x \rightarrow \infty} \frac{x^2 - 2x\mu + \mu^2}{x} = \infty,$$



$$\phi(x) \geq \sqrt{\frac{2e}{\pi}} \frac{\sqrt{\beta-1}}{\beta} e^{-\beta x^2},$$















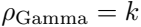










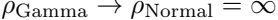














Q1 2024

2023年12月21日



$$\mathbb{E}[(S_1 + S_2)^2] = \int_0^1 (\cos(2\pi t) + \cos(2\pi t + \pi))^2 dt = \int_0^1 0 dt = 0,$$



1921 + 1921

$$= \int_0^1 \cos^2(2\pi t) dt + \int_0^1 \cos^2(2\pi t + \pi) dt$$

$$= \int_0^1 2 \cos^2(2\pi t) dt = \int_0^1 \cos(4\pi t) dt + 1 = 1.$$





$$E[S_1^2 + S_2^2] = E[S_1^2] + E[S_2^2] = E[S_1^2] + 2E[S_1 S_2] + E[S_2^2]$$

1991 + 1991 + 1991

1992-93

$$\mathbb{E}[S_1 S_2] = \int_0^1 \cos(2\pi t) \cos(2\pi t + \pi) dt = \int_0^1 -\cos^2(2\pi t) dt = -\frac{1}{2} \int_0^1 \cos(4\pi t) dt - \frac{1}{2} = -\frac{1}{2},$$

01

,

02

E

1

0

,

2

1

1

2024 + 2025

2023-04-20

$$\mathbb{E} \phi_1 \phi_2 [\mathbb{E} [S_1^2 + S_2^2]] = \mathbb{E} \phi_1 [\mathbb{E} [S_1^2]] + \mathbb{E} \phi_2 [\mathbb{E} [S_2^2]] + \mathbb{E} \phi_1 \phi_2 [\mathbb{E} [S_1 S_2]]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2$$

$$+ \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} \int_0^1 \cos(2\pi t + \phi_1) \cos(2\pi t + \phi_2) dt d\phi_1 d\phi_2$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2 = 1/2 + 1/2 = 1.$$

1992

A pixelated, black and white graphic of the text "P.O. BOX 12345". The text is rendered in a bold, blocky font with a dithered or pixelated appearance. The letters are composed of various shades of gray and black pixels. The "P" and "O" are particularly large and prominent. The "X" is formed by two intersecting lines. The numbers "12345" follow the "BOX" with a space in between. The entire graphic is set against a plain white background.



2020-2021





$${}_3F_2(1, 1, 1 + b; 2, 2; x) = \sum_{n=0}^{\infty} \frac{(1)_n (1)_n (1 + b)_n}{(2)_n (2)_n} \frac{x^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(1+b)_n}{(n+1)^2 n!} x^n = \frac{1}{b!} \sum_{n=0}^{\infty} \frac{(n+1)_b}{(n+1)^2} x^n$$

$$\underline{\underline{(a)}} \quad \frac{1}{b!} \sum_{n=0}^{\infty} \frac{\sum_{k=1}^b \begin{bmatrix} b \\ k \end{bmatrix} (n+1)^k}{(n+1)^2} x^n$$

$$\begin{aligned}
 &= \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \sum_{n=0}^{\infty} \frac{x^n}{(n+1)^{2-k}} \stackrel{(b)}{=} \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \frac{\operatorname{Li}_{2-k}(x)}{x},
 \end{aligned}$$



